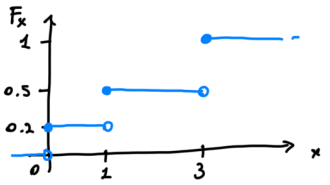
Esercizio 3. Una variabile aleatoria X ha funzione di ripartizione

$$F_X(x) = \begin{cases} 0 & x < 0 \\ 0.2 & 0 \le x < 1 \\ 0.5 & 1 \le x < 3 \\ 1 & x \ge 3. \end{cases}$$

a) Determinare il supporto e la distribuzione di probabilità di X.

b) Calcolare $\mathrm{P}(X > 0.5)$, $\mathrm{P}(1 \le X \le 3)$ e $\mathrm{P}(X < 3)$.

c) Calcolare $\mathrm{E}[X]$ e $\mathrm{var}(X)$.



$F_x(x) = \mathbb{P}(X \le x)$

a) supporto $S_x = \{0, 1, 3\}$

$\mathbb{P}(X = 0) = 0.2$

$\mathbb{P}(X = 1) = \mathbb{P}(X \le 1) - \mathbb{P}(X \le 0) =$

$= F_x(1) - F_x(0) = 0.5 - 0.2 = 0.3$

$\mathbb{P}(X = 3) = 1 - 0.5 = 0.5$

b) $\mathbb{P}(X > 0.5) = 1 - \mathbb{P}(X \le 0.5)$

$= 1 - \mathbb{P}(X = 0)$

$= 1 - 0.2 = 0.8$

$\mathbb{P}(1 \le X \le 3) = \mathbb{P}(X \le 3) - \mathbb{P}(X < 1)$

$= \mathbb{P}(X \le 3) - \mathbb{P}(X = 0)$

$= 1 - 0.2 = 0.8 = \mathbb{P}(X = 1) + \mathbb{P}(X = 3)$

$\mathbb{P}(X < 3) = \mathbb{P}(X = 0) + \mathbb{P}(X = 1) = 0.5$